

Basics in Statistics

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Regression Analysis

6 October 2026, Morning

Outline

Simple linear regression

- From comparing groups to quantifying relationships

- The regression line

- Fitted values, residuals, and R^2

- Inference for the slope

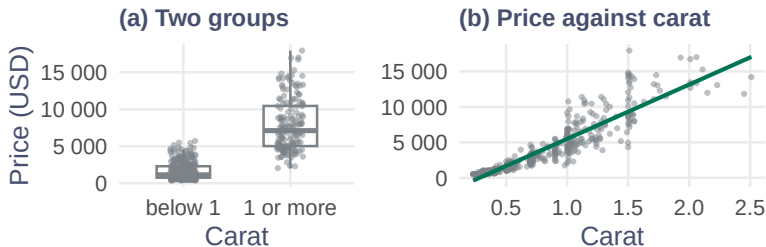
Multiple linear regression

Assumptions, diagnostics, and prediction

Model choice

Logistic regression

From group comparison to regression



500 of the 53 940 diamonds in `ggplot2::diamonds`, drawn at random; 1 carat is 0.2 g.
Video (in German) on the price of a diamond: [youtube.com/watch?v=4VF0ZA04_wM](https://www.youtube.com/watch?v=4VF0ZA04_wM)

Definition (Regression model)

A **regression model** describes how the mean of a **response** y depends on one or more **predictors** $x_1, \dots, x_p$ measured on the same study units.

The simple linear regression model

Definition (Simple linear regression)

For study units $i = 1, \dots, n$ with predictor x_i and response y_i :

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

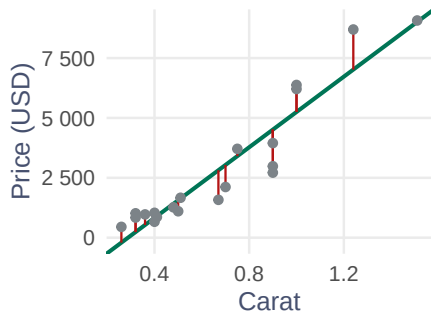
with independent errors ε_i of mean 0 and variance σ^2 . β_0 is the **intercept**, β_1 the **slope**.

Example (Diamonds)

y_i : Price of diamond i in USD; x_i : Its weight in carat. β_1 is the difference in mean price between diamonds whose weights differ by 1 carat.

The line describes the *mean* price of all diamonds of a given weight. Single diamonds scatter around it.

The least squares criterion



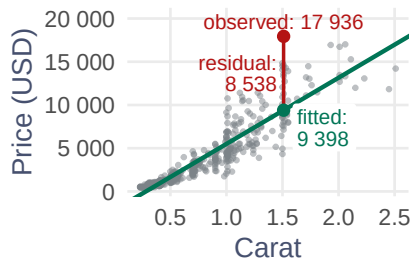
Definition (Least squares)

The **least-squares line** is the line with the smallest sum of squared vertical distances between the points and the line. Its intercept and slope are the **least-squares estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$.

Left: 20 of the 500 diamonds with their least-squares line; red: Vertical distances.

- The distances are vertical because the line predicts the price from the weight.
- Squares count deviations above and below the line alike and weight large ones more.
- In R: `lm(price ~ carat, data = d)`.

Fitted values, residuals, and R^2



500 diamonds with the least-squares line; marked: The diamond with the largest residual.

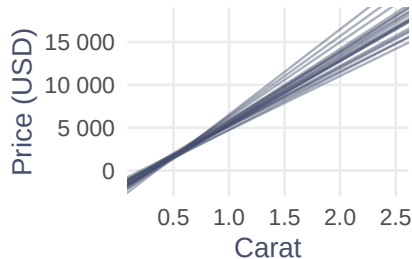
Definition (Fitted value, residual)

The **fitted value** $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ is the value of the line at x_i . The **residual** $y_i - \hat{y}_i$ is the vertical distance from the line.

500 diamonds: $\hat{y} = -2156 + 7652 \cdot \text{carat}$.

- **Residual standard error** $\hat{\sigma} = 1469$ USD: The typical distance of a price from the line.
- $R^2 = 0.84$: The share of the variation of the prices that the line reproduces.
- The intercept -2156 USD is the line at 0 carat and only fixes its height.

Uncertainty of the slope



20 samples of 30 diamonds each:
Slopes from 6300 to 10 000 USD
per carat.

Definition (Test and confidence interval)

The **standard error** $SE(\hat{\beta}_1)$ measures how much the slope varies from sample to sample.

$t = \hat{\beta}_1 / SE(\hat{\beta}_1)$ tests $H_0: \beta_1 = 0$, a mean that does not change with x . The 95 % confidence interval has the form of Unit 2: Estimate $\pm$ quantile $\cdot$ SE.

- 500 diamonds, quantile 1.965 for 498 degrees of freedom: $7652 \pm 1.965 \cdot 148$, from 7361 to 7942 USD per carat.
- The standard error shrinks with more observations and with a wider spread of the predictor.

The output of `summary(lm())`

Call:

```
lm(formula = price ~ carat, data = d)
```

Coefficients: **one row per coefficient: estimate, uncertainty, test**

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2155.6	130.7	-16.49	<2e-16	***
carat	7651.7	147.8	51.77	<2e-16	***

the model as a whole: spread, share of variation, joint test

Residual standard error: 1469 on 498 degrees of freedom

Multiple R-squared: 0.8433, Adjusted R-squared: 0.843

F-statistic: 2680 on 1 and 498 DF, p-value: < 2.2e-16

- Row carat: Slope 7651.7 USD per carat, standard error 147.8, $t = 51.77$, and the p-value of $H_0: \beta_1 = 0$. <2e-16 means below $2 \cdot 10^{-16}$.
- Below the table: $\hat{\sigma} = 1469$ USD on $n - 2 = 498$ degrees of freedom, $R^2 = 0.84$, and the F test. `confint(m)` gives the confidence intervals.
- Video (in German): youtube.com/watch?v=NEfjirp0j7s

Exercise: Reading the output

```
summary(lm(price ~ carat, data = d))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2155.6	130.7	-16.49	<2e-16 ***
carat	7651.7	147.8	51.77	<2e-16 ***

Residual standard error: 1469 on 498 degrees of freedom
Multiple R-squared: 0.8433, Adjusted R-squared: 0.843

Exercise 3.1 (3 minutes, in pairs)

Use the output above.

1. What does 1469 mean, in words and with its unit?
2. Which mean price does the line give for diamonds of 1 carat?
3. A colleague writes: “The weight explains 84 % of the price of a diamond.” Name two problems with this sentence.

Exercise: Simple regression in R

Go to `u3_regression_analysis.R`, section 3.1

R exercise 3.1 (10 minutes, in pairs)

Question: How does the price depend on the length x of a diamond (in mm)?

1. Work through section 3.1: Run every demo, then solve the task below it.
2. Report the slope with its unit and its 95 % confidence interval, and the mean price at a length of 6 mm.
3. Which mean price does the line give at 4 mm? What does that say about the straight line?

Outline

Simple linear regression

Multiple linear regression

- Adding a second predictor
- Confounding and adjustment
- Categorical predictors
- Interactions
- Collinearity
- Statistical significance and relevance

Assumptions, diagnostics, and prediction

Model choice

Logistic regression

The multiple linear regression model

Definition (Multiple linear regression)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon,$$

with errors as in simple regression. β_j is the difference in mean response between two study units that differ by one unit in x_j and agree in all other predictors of the model.

Example (Diamonds)

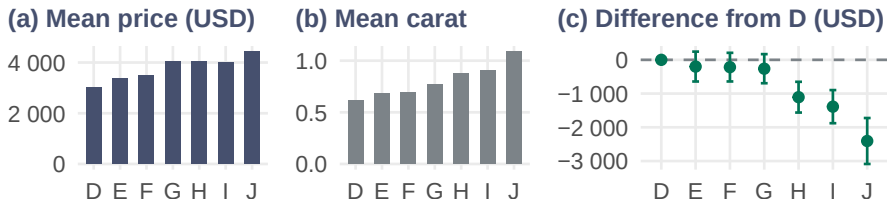
In `lm(price ~ carat + color, data = d)` the carat coefficient compares diamonds of the same colour whose weights differ by 1 carat.

- Least squares, fitted values, residuals, and R^2 work as before.
- Only the predictors in the model are held fixed. Without randomization, a coefficient describes an association, not an effect.

Confounding and adjustment

Definition (Confounder, adjustment)

A **confounder** is associated with the predictor and with the response. Including it as a predictor is called **adjustment**.



500 diamonds, colour from D (colourless) to J (tinted). (a) Without adjustment, J costs 1394 USD more than D, (b) because J diamonds are heavier. (c) At equal weight, J costs 2406 USD less ($\text{lm}(\text{price} \sim \text{carat} + \text{color})$, 95% CIs).

Dummy coding

Definition (Dummy variables)

A categorical predictor with k categories enters the model as $k - 1$ **dummy variables** with values 0 and 1. The category without a dummy is the **reference category**.

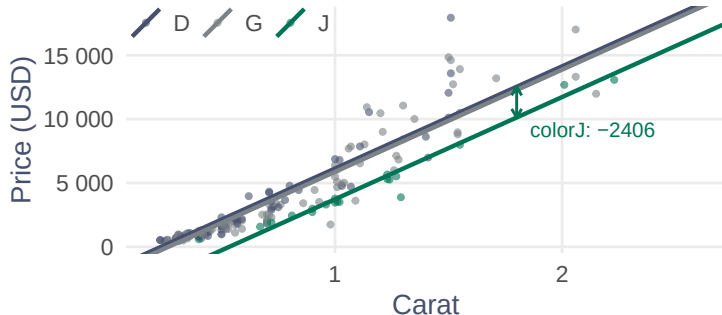
Example (Colour, seven categories)

Six dummies $d_E, d_F, \dots, d_J$; D is the reference:

colour	d_E	d_F	...	d_J
D	0	0	...	0
E	1	0	...	0
...				
J	0	0	...	1

- R codes every unordered factor this way and takes the first level as the reference.

Example: Parallel regression lines



Diamonds of colours D, G, and J with the lines of `lm(price ~ carat + color)`:
One slope, one intercept per colour. The arrow marks the coefficient of `colorJ`.

Example: Interpreting dummy coefficients

```
summary(lm(price ~ carat + color, data = d))
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1863.2	190.0	-9.808	< 2e-16	***
carat	7998.2	141.2	56.638	< 2e-16	***
colorE	-199.4	225.3	-0.885	0.377	
colorF	-216.6	216.1	-1.002	0.317	
colorG	-262.5	219.5	-1.196	0.232	
colorH	-1106.7	231.9	-4.772	2.41e-06	***
colorI	-1387.7	249.9	-5.554	4.59e-08	***
colorJ	-2405.8	346.8	-6.936	1.27e-11	***

Residual standard error: 1353 on 492 degrees of freedom
Multiple R-squared: 0.8685, Adjusted R-squared: 0.8667

- carat: One slope for all colours, 7998 USD per carat.
- colorE to colorJ: Differences in mean price from colour D *at the same weight*. E, F, and G hardly differ from D.
- At 1 carat: $-1863 + 7998 = 6135$ USD for colour D and $6135 - 2406 = 3729$ USD for colour J.

Exercise: Dummy coefficients

Exercise 3.2 (5 minutes, in pairs)

Use the output on the previous slide.

1. Which mean price does the model give for a diamond of colour H with 1.5 carat?
2. What is the difference in mean price between colours H and J at the same weight?
3. The p-value of `colorE` is 0.377. Which null hypothesis does it belong to?
4. A colleague refits the model with J as the reference category. Which numbers of the output change, and which do not?

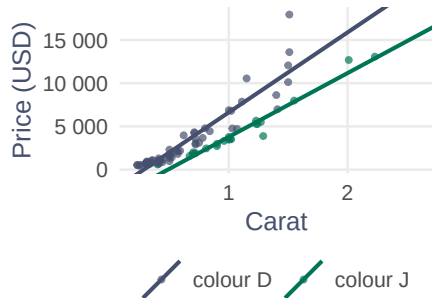
The interaction model

Definition (Interaction)

An **interaction** gives each group its own slope. With a numerical predictor x_i and a dummy variable z_i :

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i.$$

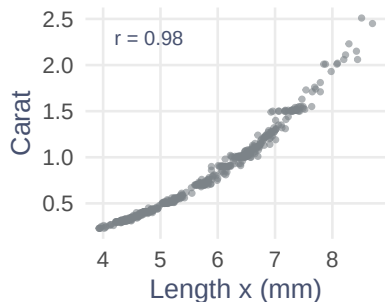
Group $z = 0$ has the slope β_1 , group $z = 1$ the slope $\beta_1 + \beta_3$.



Colours D and J with the lines of `lm(price ~ carat * color)`.

- Diamonds: The price rises by 9252 USD per carat for colour D and by 7383 USD for colour J. The difference between the colours grows with the weight.
- In R: `price ~ carat * color`. `anova()` of the models without and with the interaction gives $p < 0.001$: The slopes differ.

Collinearity



Weight against length of the 500 diamonds.

Collinearity: Two or more predictors carry almost the same information. Their coefficients are then imprecise.

Coefficients (standard errors) for the price:

predictors	carat	length x
carat	7652 (148)	
length		3048 (71)
both	9852 (726)	-925 (299)

- Weight and length carry almost the same information: With both, the standard error of carat is five times as large, and length gets a negative sign.
- **Variance inflation factor** $1/(1 - 0.98^2) = 25$. One of the two predictors is usually enough.

Example: Significance and relevance

```
summary(lm(price ~ carat + depth)), all 53940 diamonds
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4045.333	286.205	14.13	<2e-16 ***
carat	7765.141	14.009	554.28	<2e-16 ***
depth	-102.165	4.635	-22.04	<2e-16 ***

Residual standard error: 1542 on 53937 degrees of freedom
Multiple R-squared: 0.8507, Adjusted R-squared: 0.8507

- `depth` is the height of a diamond in percent of its diameter. One percentage point more depth goes with 102 USD less at the same weight, $p < 10^{-100}$.
- Without `depth`, $\hat{\sigma}$ is 1549 instead of 1542 USD: For the price, `depth` hardly matters.

Exercise: Multiple regression in R

Go to `u3_regression_analysis.R`, section 3.2

R exercise 3.2 (15 minutes, in pairs)

Response: Price. Predictors: Weight and cut (cut, from Fair to Ideal), later depth and table width.

1. Work through section 3.2: Run every demo, then solve the task below it.
2. Which cut is the cheapest without adjustment, and which at equal weight? Why?
3. Is the interaction of weight and cut needed?
4. Are weight and depth collinear?
5. Is the table width significant, and does it matter for the price?

Outline

Simple linear regression

Multiple linear regression

Assumptions, diagnostics, and prediction

- Assumptions of the linear model

- Reading the diagnostic plots

- Remedies for violated assumptions

- Prediction

Model choice

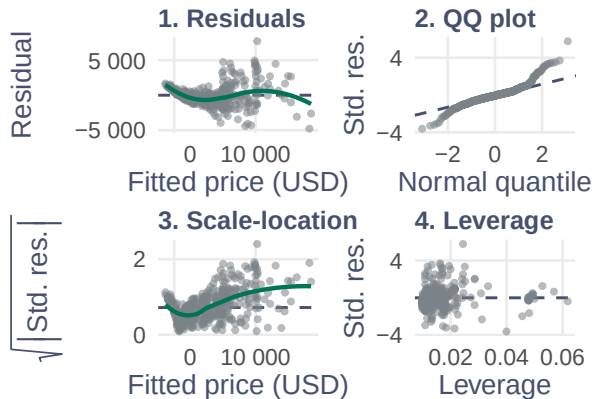
Logistic regression

Assumptions of the linear model

Assumption	Meaning	If violated
linearity	The mean of y follows the model; no term is missing.	The coefficients are systematically off; more data do not help.
independence	The errors of different units are independent.	Standard errors, p-values, and intervals are usually too small.
equal variance	The scatter around the model is the same for all fitted values.	The estimates are fine; their standard errors are not.
normality	The errors are normally distributed.	Matters for small n and for prediction intervals.

The assumptions concern the errors, not the distribution of y or of the predictors. Independence follows from the study design; no residual plot shows it.

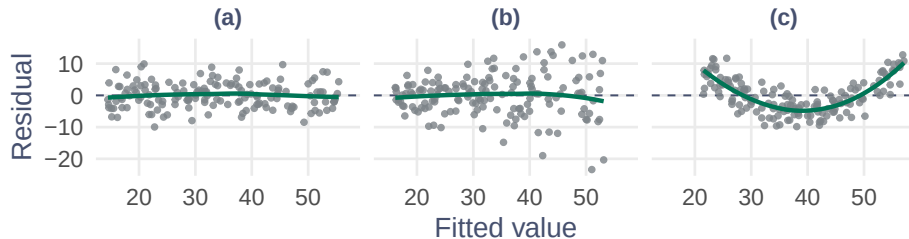
Diagnostic plots



`plot(m) for lm(price ~ carat + color).`
Dashed: Expected pattern; solid: Smoothed trend.

- Panels 1 and 3: The scatter grows with the fitted price, and the trend is curved. Equal variance and linearity are violated; 41 small diamonds get a negative price.
- A curve calls for a missing term, such as a squared term or a left-out predictor. A funnel calls for the logarithm of the response.

Exercise: Residual plots

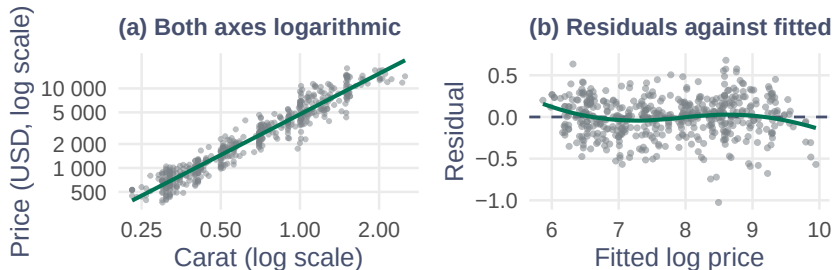


Residuals against fitted values for three simulated regressions with 150 observations each; dashed: Zero line, solid: Smoothed trend.

Exercise 3.3 (3 minutes, in pairs)

For each of the panels (a), (b), (c): Which assumption is violated, if any? What would you try first? Which results of the fit can still be used?

A log model for the diamonds



(a) 500 diamonds on logarithmic axes with the line of $\text{lm}(\log(\text{price}) \sim \log(\text{carat}))$. (b) Residuals of the colour model from the diagnostic plots, now with logarithms: $\text{lm}(\log(\text{price}) \sim \log(\text{carat}) + \text{color})$.

- On the log scale the relationship is linear, the scatter is about constant, and no fitted price is negative.

Interpretation of logarithmic models

Definition (Reading a log model)

$\log y$ on $\log x$: A 1 % larger x goes with about β_1 % more y .

$\log y$ on a dummy variable: y is multiplied by e^β , in R `exp()`.

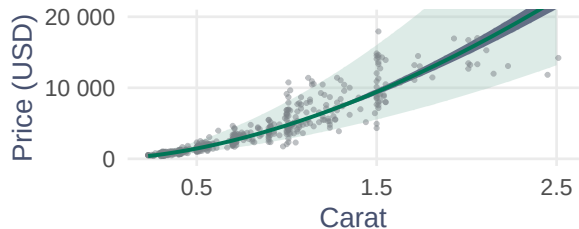
Example (`lm(log(price) ~ log(carat) + color)`)

Slope 1.76: A diamond that is 1 % heavier costs about 1.76 % more.

`colorJ` = -0.474 : $e^{-0.474} = 0.62$, so at equal weight a diamond of colour J costs 38 % less than one of colour D (95 % confidence interval 30 to 45 % less).

- Back-transformed with `exp()`, a fitted value estimates the median price, not the mean price.

Confidence interval and prediction interval



■ confidence interval ■ prediction interval

`lm(log(price) ~ log(carat))`, transformed
back with `exp()`.

Definition (Two intervals)

The **confidence interval** covers the mean response at x_0 . The **prediction interval** covers the response of one new unit at x_0 ; it also contains the scatter $\hat{\sigma}$ of single observations. Its standard error is $\sqrt{\hat{\sigma}^2 + \text{SE}(\hat{y}_0)^2}$.

- `lm(log(price) ~ log(carat))` at 1 carat: Median price 4729 USD, confidence interval 4591 to 4872 USD, prediction interval 2784 to 8034 USD.
- In R: `predict(m, newdata, interval = "confidence")` or `"prediction"`. Outside the observed range both intervals are guesses (extrapolation).

Exercise: Confidence and prediction interval

Exercise 3.4 (5 minutes, in pairs)

Log-log model without colour for 500 diamonds. For a diamond of 1 carat R returns on the log scale

$$\hat{y}_0 = 8.462, \quad \text{SE}(\hat{y}_0) = 0.015, \quad \hat{\sigma} = 0.269, \quad \text{quantile } 1.965.$$

1. Compute the 95 % confidence interval on the log scale and transform both ends back with `exp()`.
2. Do the same for the 95 % prediction interval of one diamond.
3. A dealer offers a diamond of 1 carat for 9000 USD. Is that unusual?

Exercise: Diagnostics and prediction in R

Go to `u3_regression_analysis.R`, section 3.3

R exercise 3.3 (12 minutes, in pairs)

Response: Price. Predictors: Weight and clarity (clarity, from I1, the worst, to IF, flawless).

1. Work through section 3.3: Run every demo, then solve the task below it.
2. Which assumptions are violated in `lm(price ~ carat + clarity)`? Which panels show it?
3. Log model: By which factor does the price change from I1 to IF at equal weight?
4. Give the 95 % prediction interval for a diamond of 0.5 carat and clarity VS1.

Outline

Simple linear regression

Multiple linear regression

Assumptions, diagnostics, and prediction

Model choice

Comparing models

Logistic regression

Comparing models

Definition (Criteria)

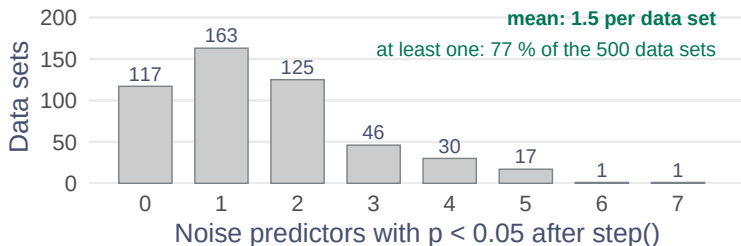
The **adjusted R^2** (larger is better) and the **AIC** (smaller is better) weigh the fit against the number of coefficients; R^2 never falls when a predictor is added. If one model contains all predictors of the other, an F test compares them: `anova(m0, m1)`.

Four models for `log(price)`, 500 diamonds:

	predictors	coefficients	adjusted R^2	AIC	$\hat{\sigma}$
m1	<code>log(carat)</code>	2	0.929	111.0	0.269
m2	<code>m1 + colour</code>	8	0.945	-15.6	0.236
m3	<code>m2 + clarity</code>	15	0.981	-535.7	0.139
m4	<code>m3 + cut</code>	19	0.982	-568.5	0.134

Every step lowers the AIC; `anova(m3, m4)`: $F = 10.2$ on 4 and 481 degrees of freedom, $p < 0.001$. Clarity improves the model much more than cut.

Stepwise selection



500 simulated data sets with $n = 100$, a response, and 20 predictors, all independent random numbers: Number of predictors with $p < 0.05$ in the model chosen by step().

- step() adds or removes predictors one at a time by the AIC. In pure noise it keeps at least one “significant” predictor in 77 % of the data sets.
- After a selection the p-values are too small and the intervals too narrow. If the aim is to explain, fix the predictors in advance from subject knowledge.

Outline

Simple linear regression

Multiple linear regression

Assumptions, diagnostics, and prediction

Model choice

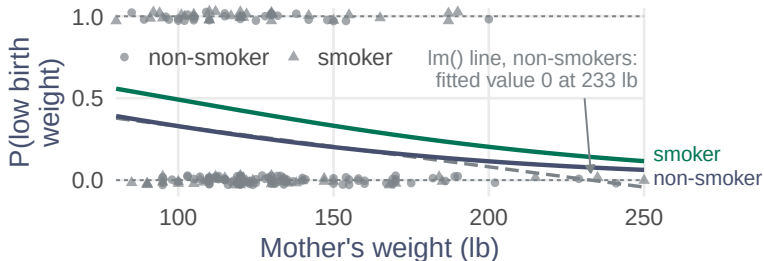
Logistic regression

- A binary response

- The logistic regression model

- Logistic regression in R

A binary response



MASS::birthwt: 189 births, 59 with a low birth weight (coded 1), against the mother's weight, by smoking status. Solid: Logistic regression; dashed: `lm()`.

- For a 0/1 response the mean is a probability, $P(y = 1) = \pi$. A straight line can leave the range from 0 to 1: `lm()` gives fitted values down to -0.020 .

Odds and odds ratios

Definition (Odds, log-odds, odds ratio)

The **odds** of an event are its probability divided by the probability that it does not happen, $\pi/(1 - \pi)$. The **log-odds** are the logarithm of the odds. The **odds ratio** of two groups is the ratio of their odds.

probability π	0.1	0.2	0.5	0.8	0.9
odds	0.11	0.25	1	4	9
log-odds	-2.20	-1.39	0	1.39	2.20

Example (Smoking and low birth weight, Unit 2)

Smokers: 30 low birth weights, 44 normal, odds $30/44 = 0.68$. Non-smokers: 29 and 86, odds 0.34. Odds ratio $0.68/0.34 = 2.02$.

Exercise: Odds and odds ratio

Exercise 3.5 (5 minutes, in pairs)

Münster Marathon 2026, net time below four hours by sex:

	below 4:00	4:00 or more	total
women	242	524	766
men	1278	1326	2604

1. Compute the proportion and the odds of a time below 4:00 for women and for men.
2. Compute the odds ratio, men against women.
3. “Men are twice as likely to finish below four hours.” Compare with the ratio of the two proportions.

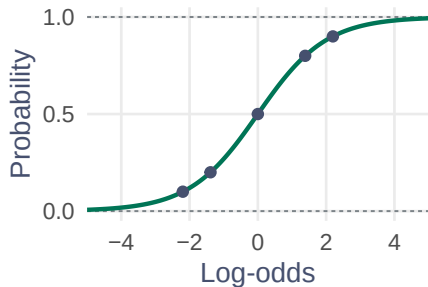
The logistic regression model

Definition (Logistic regression)

For independent binary responses with $P(y = 1) = \pi$:

$$\log \frac{\pi}{1 - \pi} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

The coefficients are estimated by maximum likelihood (Unit 2).



The logistic curve turns any log-odds into a probability between 0 and 1. Points: The table of the odds frame.

- e^{β_j} is the odds ratio for one unit more of x_j , the other predictors fixed.
- In R: `glm(low ~ smoke + lwt, family = binomial, data = bw)`.

Example: The output of glm()

```
— glm(low ~ smoke + lwt, family = binomial, data = bw), lwt: mother's weight in pounds —  
              Estimate Std. Error z value Pr(>|z|)  
(Intercept)  0.62200    0.79592   0.781   0.4345  
smokeyes     0.67667    0.32470   2.084   0.0372 *  
lwt         -0.01332    0.00609  -2.188   0.0287 *  
Null deviance: 234.67  on 188  degrees of freedom  
Residual deviance: 224.34  on 186  degrees of freedom  
AIC: 230.34
```

- smokeyes = 0.677 on the log-odds scale; $e^{0.677} = 1.97$ (95 % confidence interval 1.04 to 3.73): Smokers have about twice the odds of a low birth weight, at the same weight of the mother.
- lwt: $e^{-0.0133} = 0.987$ per pound, $e^{-0.133} = 0.875$ per 10 pounds.
- Read z value like the t value of lm(). For the deviance and the AIC below the table, smaller is better.

Predicted probabilities

Example (Predicted probability of a low birth weight)

First the log-odds from the coefficients, then the probability. Smoker, 130 pounds:

$$0.622 + 0.677 - 0.01332 \cdot 130 = -0.433, \quad \frac{e^{-0.433}}{1 + e^{-0.433}} = 0.393.$$

Non-smoker, 130 pounds: 0.248.

- `predict(g, newdata, type = "response")` returns probabilities; without `type` it returns log-odds.
- Smokers have 1.59 times the probability (0.393 against 0.248) but 1.97 times the odds: An odds ratio is not a ratio of probabilities.

Exercise: Logistic regression in R

Go to `u3_regression_analysis.R`, section 3.4

R exercise 3.4 (10 minutes, in pairs)

Response: Low birth weight (`low`). Predictors: Smoking, the mother's weight, and hypertension (`ht`, 1 = yes).

1. Work through section 3.4: Run every demo, then solve the task below it.
2. Report the odds ratio of hypertension with its 95 % confidence interval.
3. Give the probability of a low birth weight for a smoker of 120 pounds without and with hypertension.